Assignment 3 — React.js

Subject: Web Development
Assignment No.: 03
Title: Building a Student Interface Using React.js
Difficulty: Intermediate
Submission Format: PDF / DOCX

Objective

To understand React.js and develop a simple component-based student interface.

Problem Statement

Modern web applications require reusable and interactive user interfaces. React.js allows developers to build interfaces using reusable components.

Questions

1. What is React.js?
2. Explain the major features of React.js.
3. What is a React component?
4. Explain functional components with an example.
5. What is JSX?
6. Explain props in React.
7. What is state?
8. Explain the `useState()` Hook.
9. What is `useEffect()`?
10. Explain event handling in React.
11. Explain conditional rendering.
12. Explain rendering lists using `map()`.
13. What is React Router?
14. Explain the advantages of component-based development.